



AUTONOMOUS SYSTEMS

PROJECTS 2025/26

Instituto Superior Técnico

Departamento de Engenharia Electrotécnica e de Computadores

April 2026



LIST OF AVAILABLE REAL ROBOTS AND DEVICES

Pioneer 3DX (7 units)



AlphaBot2 with camera (many units)



Turtlebot3 with RPLIDAR and camera (4 units)



Hokuyo URG-04LX-UG01 (5 units)



Turtlebot4 with RPLIDAR and camera (5 units)





PROJECT TOPICS

Project topics and code scheme:

- [L__] – Localization:** estimate in real-time the pose (position+orientation) of a mobile robot; evaluate estimation accuracy, as well as absolute localization, and robustness to “kidnapping”.
- [M__] – Mapping:** estimate the map of the environment using Occupancy Grid Mapping; evaluate quality of the map with respect to the ground truth.
- [S__] – Simultaneous Localization And Mapping (SLAM):** estimate simultaneously the trajectory (position+orientation) of a mobile robot and the landmark positions (map); evaluate estimation accuracy of both trajectory and landmarks.
- [DMA] – Decision Making:** solve a maze using MDPs/RL and markers to identify the goal and help odometry in localizing the robot



SENSOR CHOICE RECOMMENDATIONS

Localization: fuses relative and absolute sensing:

- for relative sensing, use wheel odometry
- for absolute sensing, prefer Laser or camera (natural landmarks), while sonar is challenging

Mapping: registers distance measurements given known localization

- may use AMCL (from ROS) or well-calibrated odometry
- for distance sensor, may use Laser, depth camera (Kinect) or sonars (challenging)

Simultaneous Localization And Mapping (SLAM): fuses landmark measurements with relative sensing:

- for relative sensing, use wheel odometry
- use fiducials markers, e.g., camera with ARuCO or AprilTag markers

Decision Making (DM): moves in a maze using the optimal policy for a given MDP (in a later stage it can learn it model free using reinforcement learning) and uses fiducial markers (Alphabot camera)



LIST OF TOPICS (REAL ROBOTS)

		Pioneer 3DX	Turtlebot3&4	AlphaBot2
<u>L</u> ocalization	Extended <u>K</u> alman Filter (EKF)	LKP	LKT	-
	<u>M</u> onte Carlo Localization (MCL)	LMP	LMT	-
<u>M</u> apping	Occupancy Grid Mapping	MP	MT	-
<u>S</u> LAM	E <u>K</u> F-SLAM	SKP	SKT	-
	<u>E</u> astSLAM	SFP	SFT	-
<u>D</u> ecision-Making	MDP/RL	-	-	DMA



PROJECT

11 PROJECT TYPOLOGIES:

- **Groups of 4 students**
 - **Using real robots and sensors**
 - **Validated and evaluated using collected data**
 - **Students are suggested to spread over all projects**
- typologies (all with same level of difficulty)**



PROJECT TIPS

- **Solid theoretical background**
 - formalize the problem, but do not write a tutorial
 - explain the algorithm, not the code
- **Develop and validate your algorithm on a micro-simulator**
- **Test as soon as possible:** it is better to test partial implementations early, than postponing to when everything is implemented
- **Avoid running your algorithm in real time:** instead, datasets should be used (e.g., rosbags), for work productivity and repeatability
- **Thorough experimental results**
 - try a variety of experimental conditions
 - for each one, run multiple times and analyze statistically
- **Objective analysis of the results is more important than “just working”**



PROJECT TIPS

What is a micro-simulator?

- simple simulator in Python
- use simple environments (e.g., mapping of inside a rectangle)
- re-use the same motion and measurement models for the project to generate synthetic data
 - start with an initial pose
 - generate next poses using motion model
 - generate measurements using measurement model
- use this micro-simulator to develop and test your code!

Why a micro-simulator?

- no dependency on neither ROS nor access to real robots
- if the project works for synthetic data, it should in principle work for real data! (you are half there)



PROJECT ASSESSMENT AND SCHEDULE (1)

- **Continuous assessment:** *each group does an oral progress presentation (1 group member per presentation) every week in its designated slot (4 groups per slot) – total of 5 intermediate presentations per group, max 20 minutes including presentation (10-15 mins) and Q&A (slides not mandatory but highly recommended)*
- **Project progress presentations during laboratory sessions** *start on 27 April (second week of classes)*
- *Projects presented by faculty on 20 April in the theoretical classes*

Project report and code deadline:

5/Jun/2026

(6 page IEEE paper template)

Project discussions:

8-9/Jun/2026



PROJECT ASSESSMENT AND SCHEDULE (2)

	Seg	Ter	Qua	Qui	Sex
8:00	T1	T2			
8:30			L 7		L10
9:00	IA	IA	LSDC1		LSDC1
9:30	PL	PL	AF		AF
10:00					L11
10:30	L1	L3			LSDC1
11:00	LSDC1	LSDC1		L9	JL
11:30	JM	RV		LSDC1	L12
12:00	L2	L4		AF	LSDC1
12:30	LSDC1	LSDC1			JL
13:00	JM	RV			L4.1
13:30	L4.2				LSDC1
14:00	LSDC1				RV
14:30					
15:00		L5			
15:30		LSDC1	L 8		
16:00		RV	LSDC1		
16:30		L6	AF		
17:00		LSDC1			
17:30		RV			
18:00					
18:30					
19:00					
19:30					
20:00					



PROJECT ASSESSMENT AND SCHEDULE (3)

	Session				
Slot	1	2	3	4	5
1	27-April	4-May	11-May	18-May	25-May
2	27-April	4-May	11-May	18-May	25-May
3	28-April	5-May	12-May	19-May	26-May
4	28-April	5-May	12-May	19-May	26-May
5	28-April	5-May	11-May	18-May	25-May
6	28-April	5-May	12-May	19-May	26-May
7	29-April	6-May	13-May	20-May	27-May
8	29-April	6-May	13-May	20-May	27-May
9	30-April	7-May	13-May	20-May	27-May
10	1-May	8-May	15-May	22-May	29-May
11	1-May	8-May	14-May	21-May	28-May
12	1-May	8-May	15-May	22-May	29-May



PROJECT ASSESSMENT AND SCHEDULE (3)

Project Grading:

- **FAIL:** *nothing works, not much relevant work done in design + implementation, no reasonable explanation for failure to show results*
- **10-14:** *at least some experimental results can be shown, significant design + implementation work made of at least fair quality*
- **15-17:** *good experimental results, significant design + implementation work made of at least good quality and supported by theory*
- **18-19:** *very good experimental results and design + implementation work made and supported by theory*
- **20:** *excellent and flawless experimental results and design + implementation work made and supported by theory; in exceptional cases could correspond to the factors listed for 18-19, extended with some original unsolicited extra work*



WHAT'S NEXT

Hands-on sessions with ROS2 on **first week**

- Each shift uses 2 assigned lab slots (all shift students go to both slots)
- Prepare for these sessions by reading the “**Laboratory guide**” slides
- First slot in **LSDC1** and second slot in **LSDC4**

1. [**from 20 April 17:00 to 22 April 17:00**] Choose the project for your group using a web link that will be made available at the course webpage

2. Group start preparing the first presentation, where we expect:

- i) problem statement
- ii) literature readings
- iii) work planning (see next slide)



Workplan suggestion

Session 1: Project presentation - problem statement, readings, workplan

Session 2: Explain by own words the algorithm. Get, visualize and represent real robot sensor data

Session 3: Develop and validate the project in a micro-simulator

Session 4: Integrate with ROS and validate project in real robot data

Session 5: Systematic and comparative (w.r.t. baseline) experimentation with quantitative analysis, that is:

- same environment/trajectory, multiple runs
- a variety of environments/trajectories